Chapter 9. Epilogue—Life Outside $\mathbb{R}^n$

Differentiable manifolds and Riemannian manifolds

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Overview

We have dealt with submanifolds of euclidean space and forms on open sets of euclidean space: conceptually simple, but important ideas are sometimes obscured by the familiar surroundings, and in mathematics and physics manifolds often occur as abstract spaces. This chapter sketches how to move outside $\mathbb{R}^n$.

§41. Differentiable Manifolds and Riemannian Manifolds

Motivation. So far manifolds sat inside $\mathbb{R}^n$. Abstract manifolds remove the ambient space and keep only coordinate charts. Everything proved for submanifolds, including forms and Stokes' theorem, carries over once tangent vectors are defined intrinsically.

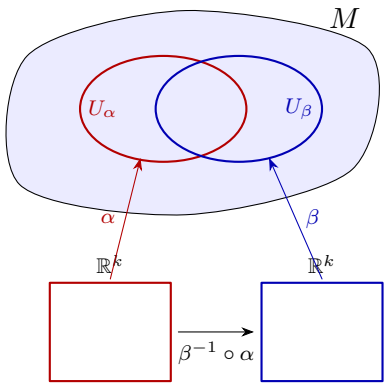


Figure. *Coordinate patches α, β of an abstract manifold overlap; the transition map $\beta^{-1}\circ\alpha$ between open sets of $\mathbb{R}^k$ must be C^r. Smoothness on M is defined through these charts, without any ambient $\mathbb{R}^n$.*

41.1 Differentiable manifolds

Definition 41.1 (differentiable manifold)

Let M be a metric space. Suppose there is a collection of homeomorphisms $\alpha_i : U_i \to V_i$, where U_i is open in $\mathbb{H}^k$ or $\mathbb{R}^k$ and V_i is open in M, such that the V_i cover M. (α_i is a **homeomorphism** if it carries U_i onto V_i one-to-one and both α_i, α_i^{-1} are continuous.) Suppose the α_i overlap with class C^∞: the **transition function** $\alpha_i^{-1}\circ\alpha_j$ is of class C^∞ whenever $V_i \cap V_j$ is non-empty. The α_i are **coordinate patches** on M, as is any other homeomorphism $\alpha : U \to V$ (U open in $\mathbb{H}^k$ or $\mathbb{R}^k$, V open in M) that overlaps the α_i with class C^∞. M with this collection of patches is a **differentiable k-manifold** (of class C^∞).

For $k = 1$ we again allow domains open in $\mathbb{L}^1$. If there is a patch $\alpha : U \to V$ about p with U open in $\mathbb{R}^k$, p is an **interior point**; otherwise a **boundary point**; the boundary points form ∂M. If $\alpha : U \to V$ is a patch about p, then $p \in \partial M$ iff U is open in $\mathbb{H}^k$ and $p = \alpha(\mathbf{x})$ for some $\mathbf{x} \in \mathbb{R}^{k-1} \times 0$ (proof as in Lemma 24.2). Throughout, M denotes a differentiable k-manifold.

Definition 41.2 (orientation)

Patches α_0, α_1 on M **overlap positively** if $\det D(\alpha_1^{-1}\circ\alpha_0) > 0$. If M can be covered by patches overlapping positively, M is **orientable**. An **orientation** of M consists of such a covering together with all other patches overlapping these positively; an **oriented manifold** is a manifold with an orientation. If $\{\alpha_i\}$ is an orientation, $\{\alpha_i \circ r\}$ (r the reflection of $\mathbb{R}^k$) is

the **opposite** orientation.

If $\partial M \neq \emptyset$, then ∂M is a differentiable $(k-1)$-manifold without boundary, with patches $\alpha \circ b$ (α a patch about $p \in \partial M$, $b(x_1, \ldots, x_{k-1}) = (x_1, \ldots, x_{k-1}, 0)$), as in Theorem 24.3. If α_0, α_1 overlap positively, so do $\alpha_0 \circ b$ and $\alpha_1 \circ b$ (Theorem 34.1). So if M is oriented, ∂M can be oriented by restricting the patches of the orientation of M; for k even this is the **induced orientation** of ∂M, for k odd its opposite is.

Definition 41.3 (C^∞ maps between manifolds)

Let M, N be differentiable manifolds of dimensions k, n, $A \subset M$ and $f : A \to N$. f is **of class** C^∞ if for each $x \in A$ there are a patch $\alpha : U \to V$ on M about x and a patch $\beta : W \to Y$ on N about $y = f(x)$ such that $\beta^{-1} \circ f \circ \alpha$ is of class C^∞ as a map of a subset of $\mathbb{R}^k$ into $\mathbb{R}^n$. (Independent of the patches, since transition functions are C^∞.) A one-to-one $f : M \to N$ onto N with f and f^{-1} of class C^∞ is a **diffeomorphism**.

41.2 Tangent vectors

Without a surrounding euclidean space, what is a tangent vector? For M in $\mathbb{R}^n$, a tangent vector at p is the velocity $(p; D\gamma(t_0))$ of a C^∞ curve $\gamma : [a, b] \to M$ with $\gamma(t_0) = p$. For abstract M we cannot differentiate γ, but for a patch $\alpha : U \to V$ about p, $\alpha^{-1} \circ \gamma$ maps a subset of $\mathbb{R}^1$ into $\mathbb{R}^k$, so we may think of the "derivative" of γ at t_0 as the function $\mathbf{v}$ assigning to each patch α about p the matrix $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$. If α_0, α_1 are two patches about p, the chain rule gives

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \qquad g = \alpha_1^{-1} \circ \alpha_0, \quad \mathbf{x}_0 = \alpha_0^{-1}(p).$$

Definition 41.4 (tangent vector, tangent space)

Given $p \in M$, a **tangent vector** to M at p is a function $\mathbf{v}$ assigning to each coordinate patch $\alpha : U \to V$ about p a k by 1 column matrix $\mathbf{v}(\alpha)$, such that for any two patches α_0, α_1 about p,

$$\mathbf{v}(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{v}(\alpha_0), \tag{$*$}$$

where $g = \alpha_1^{-1} \circ \alpha_0$ and $\mathbf{x}_0 = \alpha_0^{-1}(p)$. The entries of $\mathbf{v}(\alpha)$ are the **components** of $\mathbf{v}$ with respect to α.

By $(*)$, $\mathbf{v}$ is determined by its components in a single patch, and we may define $(a\mathbf{v} + b\mathbf{w})(\alpha) = a\mathbf{v}(\alpha) + b\mathbf{w}(\alpha)$. The tangent vectors at p form the **tangent space** $\mathcal{T}_p(M)$, a k-dimensional vector space: if $\alpha(\mathbf{x}) = p$, the map $\mathbf{v} \mapsto (\mathbf{x}; \mathbf{v}(\alpha))$ is a linear isomorphism of $\mathcal{T}_p(M)$ onto $\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k)$, whose inverse is denoted $\alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$; so $\alpha_*(\mathbf{x}; \mathbf{v}(\alpha)) = \mathbf{v}$. For a C^∞ curve $\gamma : [a, b] \to M$ with $\gamma(t_0) = p$, the **velocity vector** is defined by $\mathbf{v}(\alpha) = D(\alpha^{-1} \circ \gamma)(t_0)$; every tangent vector at p is the velocity vector of some curve.

Tangent vectors as derivations

A common alternative: to $\mathbf{v} \in \mathcal{T}_p(M)$ associate an operator $X_{\mathbf{v}}$ on real-valued C^∞ functions near p, the **derivative with respect to** $\mathbf{v}$. If $\mathbf{v}$ is the velocity of γ at t_0 and $\alpha(\mathbf{x}) = p$, then writing $f \circ \gamma = (f \circ \alpha) \circ (\alpha^{-1} \circ \gamma)$,

$$\frac{d(f \circ \gamma)}{dt}(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot D(\alpha^{-1} \circ \gamma)(t_0) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha),$$

which depends only on f and $\mathbf{v}$. So define $X_{\mathbf{v}}(f) = D(f \circ \alpha)(\mathbf{x}) \cdot \mathbf{v}(\alpha)$; it is independent of α, $X_{\mathbf{v}+\mathbf{w}} = X_{\mathbf{v}} + X_{\mathbf{w}}$, $X_{c\mathbf{v}} = cX_{\mathbf{v}}$, and for $M = \mathbb{R}^k$ it is the directional derivative of f with

respect to $\mathbf{v}$. It satisfies (1) *locality*: if f, g agree near p, $X_{\mathbf{v}}(f) = X_{\mathbf{v}}(g)$;
(2) *linearity*: $X_{\mathbf{v}}(af + bg) = aX_{\mathbf{v}}(f) + bX_{\mathbf{v}}(g)$;
(3) the *product rule*: $X_{\mathbf{v}}(f \cdot g) = X_{\mathbf{v}}(f)g(p) + f(p)X_{\mathbf{v}}(g)$. These characterize $X_{\mathbf{v}}$: every operator X on C^∞ functions near p satisfying (1)–(3) equals $X_{\mathbf{v}}$ for a unique $\mathbf{v} \in \mathcal{T}_p(M)$. So one can define a tangent vector at p to be such an operator; many authors prefer this "intrinsic" definition, which does not involve coordinate patches explicitly.

41.3 Forms on M

Definition 41.5 (forms on a manifold)

An **l-form** on M is a function ω assigning to each $p \in M$ an alternating l-tensor $\omega(p) \in \mathcal{A}^l(\mathcal{T}_p(M))$. If $\alpha : U \to V$ is a patch about p with $\alpha(\mathbf{x}) = p$, let $T = \alpha_* : \mathcal{T}_{\mathbf{x}}(\mathbb{R}^k) \to \mathcal{T}_p(M)$ and $T^* : \mathcal{A}^l(\mathcal{T}_p(M)) \to \mathcal{A}^l(\mathcal{T}_{\mathbf{x}}(\mathbb{R}^k))$; define $(\alpha^*\omega)(\mathbf{x}) = T^*(\omega(p))$. ω is **of class C^∞ near** p if $\alpha^*\omega$ is of class C^∞ near $\mathbf{x}$ (independent of α), and of class C^∞ if this holds for every patch.

Henceforth all forms are C^∞. $\Omega^l(M)$ is the space of l-forms on M. There are no elementary forms on M giving a canonical form, but $\alpha^*\omega = \sum_{[I]} f_I \, dx_I$ in $\mathbb{R}^k$; the f_I are the **components** of ω with respect to α.

Definition 41.6 (differential)

For an l-form ω on M, $d\omega$ is defined as follows: given $p \in M$ and tangent vectors $\mathbf{v}_1, \ldots, \mathbf{v}_{l+1}$ at p, choose a patch α about p with $\alpha(\mathbf{x}) = p$ and set

$$d\omega(p)(\mathbf{v}_1, \ldots, \mathbf{v}_{l+1}) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{v}_1(\alpha)), \ldots, (\mathbf{x}; \mathbf{v}_{l+1}(\alpha))).$$

That is, pull ω back to $\alpha^*\omega$ in $\mathbb{R}^k$, pull the $\mathbf{v}_i$ back to tangent vectors in $\mathbb{R}^k$, and apply d in $\mathbb{R}^k$. Equivalently (with $\mathbf{a}_i = \mathbf{v}_i(\alpha)$), $d\omega(p)(\alpha_*(\mathbf{x}; \mathbf{a}_1), \ldots, \alpha_*(\mathbf{x}; \mathbf{a}_{l+1})) = d(\alpha^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{a}_1), \ldots, (\mathbf{x}; \mathbf{a}_{l+1}))$, i.e. $d\omega$ is the unique $(l+1)$-form on M such that

$$\alpha^*(d\omega) = d(\alpha^*\omega)$$

for every coordinate patch α on M. (The d on the right is the operator in $\mathbb{R}^k$.)

This is independent of the patch, and $d\omega$ is C^∞.

Theorem 41.1

Let M be a compact differentiable manifold. Given a covering of M by coordinate patches, there exist functions $\phi_i : M \to \mathbb{R}$ of class C^∞, for $i = 1, \ldots, l$, such that:
(1) $\phi_i(p) \geq 0$ for each $p \in M$.
(2) For each i, the set Support ϕ_i is covered by one of the given coordinate patches.
(3) $\sum \phi_i(p) = 1$ for each $p \in M$.

Proof. Given $p \in M$, choose a patch $\alpha : U \to V$ of the covering about p, $\alpha(\mathbf{x}) = p$, and a non-negative C^∞ function $f : U \to \mathbb{R}$ whose support is compact and contained in U, with $f(\mathbf{x}) > 0$. Define $\psi_p : M \to \mathbb{R}$ by $\psi_p(y) = f(\alpha^{-1}(y))$ for $y \in V$ and $\psi_p(y) = 0$ otherwise. Since $f \circ \alpha^{-1}$ vanishes outside a compact subset of V, ψ_p is C^∞ on M. ψ_p is positive on an open set U_p about p. Cover M by finitely many U_p, for $p = p_1, \ldots, p_l$, and set $\lambda = \sum_{j=1}^{l} \psi_{p_j}$ and $\phi_i = (1/\lambda)\psi_{p_i}$. ∎

Definition 41.7 (integral of a form)

Let M be a compact oriented differentiable k-manifold and ω a k-form on M. If Support ω lies in a single patch $\alpha : U \to V$ of the orientation, $\int_M \omega = \int_{\operatorname{Int} U} \alpha^*\omega$. In general, with $\phi_1, \dots, \phi_l$ as in Theorem 41.1,

$$\int_M \omega = \sum_{i=1}^{l} \Big[\int_M \phi_i \omega \Big].$$

The usual argument shows this is well-defined and linear.

Theorem 41.2 (Stokes' theorem)

Let M be a compact, oriented differentiable k-manifold. Let ω be a $k-1$ form on M. If ∂M is non-empty, give ∂M the induced orientation; then

$$\int_M d\omega = \int_{\partial M} \omega.$$

If ∂M is empty, then $\int_M d\omega = 0$.

Proof. The proof of Theorem 37.2 goes through verbatim: all computations were carried out within coordinate patches, so no changes are needed. The special conventions for $k = 1$ and ∂M a 0-manifold are handled as before (Theorem 37.4). ■

Chapter 8 also generalizes: the **deRham group** $H^k(M)$ is the quotient of the closed k-forms on M by the exact ones, and there are methods to compute it, including a general *Mayer–Vietoris theorem* relating the deRham groups of $M = U \cup V$, U, V and $U \cap V$ (see Bott–Tu). $H^k(M)$ is obviously a diffeomorphism invariant of M. Strikingly, it is also a *topological* invariant: if M and N are homeomorphic, $H^k(M)$ and $H^k(N)$ are linearly isomorphic. This follows from *deRham's theorem*: the algebra of closed forms modulo exact forms on M is isomorphic to an algebra defined in algebraic topology for any topological space, the "cohomology algebra of M with real coefficients".

41.4 Riemannian manifolds

Not everything generalizes so readily: the volume of M and $\int_M f\, dV$ do not make sense on an abstract differentiable manifold. By §36, the volume of a compact oriented k-manifold M in $\mathbb{R}^n$ is $\mathrm{v}(M) = \int_M \omega_V$, where the volume form ω_V has value 1 on *any orthonormal basis* of $\mathcal{T}_p(M)$ in its natural orientation. This uses the inner product that $\mathcal{T}_p(M) \subset \mathcal{T}_p(\mathbb{R}^n) = p \times \mathbb{R}^n$ inherits from the dot product of $\mathbb{R}^n$; an abstract $\mathcal{T}_p(M)$ has none, so "orthonormal" is meaningless. We need an inner product on each tangent space.

Definition 41.8 (Riemannian metric)

A **Riemannian metric** on a differentiable k-manifold M is an inner product $\langle \mathbf{v}, \mathbf{w} \rangle$ on each tangent space $\mathcal{T}_p(M)$, required to be of class C^∞ as a 2-tensor field on M. A **Riemannian manifold** is a differentiable manifold together with a Riemannian metric.

("Metric" here has nothing to do with "metric space".) Every differentiable manifold has a Riemannian metric (the proof uses a partition of unity), but it is not unique. A Riemannian metric gives a volume function $V(\mathbf{v}_1, \dots, \mathbf{v}_k)$ on k-tuples in $\mathcal{T}_p(M)$ (as in the exercises of §21), so one defines, for continuous f supported in a single patch $\alpha : U \to V$,

$$\int_M f\, dV = \int_{\operatorname{Int} U} (f \circ \alpha)\, V(\alpha_*(\mathbf{x}; \mathbf{e}_1), \dots, \alpha_*(\mathbf{x}; \mathbf{e}_k)),$$

in general via a partition of unity as in §25, and $\mathrm{v}(M) = \int_M dV$. If M is compact, oriented and Riemannian, $\int_M \omega = \int_M \lambda\, dV$ where $\lambda(p)$ is the value of $\omega(p)$ on an orthonormal k-tuple in $\mathcal{T}_p(M)$ belonging to its natural orientation; if $\lambda \equiv 1$, ω is the **volume form** ω_V of M, and $\mathrm{v}(M) = \int_M \omega_V$.

Many questions arise: the *length* of a smooth curve $\gamma : [a,b] \to M$ is $\int_{t=a}^{t=b} \|\gamma_*(t;\mathbf{e}_1)\|$, the integral of the norm of its velocity vector (using the inner product on $\mathcal{T}_p(M)$); one studies *geodesics* (curves of minimal length joining two points), *curvature*, and so on—the subject of *Riemannian geometry*.

What does not generalize

Most of this book generalizes to abstract differentiable or Riemannian manifolds. The interpretation of Stokes' theorem in terms of scalar and vector fields (§38) does not: the "translation functions" of §31 depend on forms defined in $\mathbb{R}^n$, grad and div apply only to fields in $\mathbb{R}^n$, curl only in $\mathbb{R}^3$, and even the "normal vector" to a manifold depends on the surrounding space. Manifolds, differential forms and Stokes' theorem have meaning outside euclidean space; classical vector analysis does not.

Selected exercises

Motivation. The section above states several facts about abstract manifolds without proof; these exercises from Munkres' §41 supply the proofs, each by working in coordinate patches.

Exercise 41.1 (tangent vectors are velocities)

Show that if $\mathbf{v} \in \mathcal{T}_p(M)$, then $\mathbf{v}$ is the velocity vector of some C^∞ curve γ in M passing through p.

Solution.

Step 1. Choose a patch $\alpha : U \to V$ about p, $\alpha(\mathbf{x}) = p$, and let $\mathbf{a} = \mathbf{v}(\alpha)$. Since U is open in $\mathbb{R}^k$ or in $\mathbb{H}^k$, there is $r > 0$ such that every point of $\mathbb{H}^k$ within distance r of $\mathbf{x}$ lies in U (every point of $\mathbb{R}^k$, if U is open in $\mathbb{R}^k$). Let $\epsilon = r/(1 + \|\mathbf{a}\|)$.

Step 2. Choose the parameter interval J; in each case $\|t\mathbf{a}\| \le \epsilon\|\mathbf{a}\| < r$ for $t \in J$. If U is open in $\mathbb{R}^k$, let $J = [-\epsilon, \epsilon]$; then $\mathbf{x} + t\mathbf{a} \in U$. If U is open in $\mathbb{H}^k$, it suffices that the last coordinate $x_k + ta_k$ be ≥ 0 on J:

if $x_k > 0$, take $J = [-\epsilon', \epsilon']$ with $\epsilon' = \min\{\epsilon,\, x_k/(1 + |a_k|)\}$, so that $x_k + ta_k > x_k - x_k = 0$;

if $x_k = 0$, take $J = [-\epsilon, \epsilon]$, $[0, \epsilon]$ or $[-\epsilon, 0]$ according as $a_k = 0$, $a_k > 0$ or $a_k < 0$.

Step 3. Define $\gamma : J \to M$ by $\gamma(t) = \alpha(\mathbf{x} + t\mathbf{a})$. Then $\gamma(0) = p$, and $\alpha^{-1} \circ \gamma(t) = \mathbf{x} + t\mathbf{a}$ is C^∞, so γ is a C^∞ curve. Its velocity vector $\mathbf{w}$ at $t_0 = 0$ has

$$\mathbf{w}(\alpha) = D(\alpha^{-1} \circ \gamma)(0) = \mathbf{a} = \mathbf{v}(\alpha)$$

(a one-sided derivative if 0 is an endpoint of J). By condition $(*)$ of Definition 41.4, a tangent vector is determined by its components in one patch, so $\mathbf{w} = \mathbf{v}$. (For $k = 1$ and a patch on $\mathbb{L}^1$ the same argument applies with the inequalities reversed.) ■

Exercise 41.3 ($d\omega$ is well-defined)

If ω is an l-form on M, show that $d\omega$ in Definition 41.6 is independent of the choice of the coordinate patch α.

Solution. Let $\alpha_0 : U_0 \to V_0$ and $\alpha_1 : U_1 \to V_1$ be patches about p with $\alpha_0(\mathbf{x}_0) = p = \alpha_1(\mathbf{x}_1)$. Let $W = \alpha_0^{-1}(V_0 \cap V_1)$ and $g = \alpha_1^{-1} \circ \alpha_0 : W \to U_1$; g is C^∞ and $g(\mathbf{x}_0) = \mathbf{x}_1$.

Step 1. ($\alpha_{0*} = \alpha_{1*} \circ g_*$.) Let $\mathbf{x} \in W$, $\mathbf{y} = g(\mathbf{x})$, $\mathbf{a} \in \mathbb{R}^k$, and let $\mathbf{v} = \alpha_{0*}(\mathbf{x}; \mathbf{a})$, i.e. $\mathbf{v}(\alpha_0) = \mathbf{a}$. By $(*)$ of Definition 41.4, $\mathbf{v}(\alpha_1) = Dg(\mathbf{x}) \cdot \mathbf{a}$, so

$$\alpha_{0*}(\mathbf{x}; \mathbf{a}) = \alpha_{1*}(\mathbf{y}; Dg(\mathbf{x}) \cdot \mathbf{a}) = \alpha_{1*}(g_*(\mathbf{x}; \mathbf{a})).$$

Step 2. ($\alpha_0^*\omega = g^*(\alpha_1^*\omega)$ on W.) For tangent vectors $(\mathbf{x}; \mathbf{a}_1), \dots, (\mathbf{x}; \mathbf{a}_l)$, Definition 41.5 and Step 1 give

$$(\alpha_0^*\omega)(\mathbf{x})((\mathbf{x}; \mathbf{a}_i))_i = \omega(p')(\alpha_{1*}(g_*(\mathbf{x}; \mathbf{a}_i)))_i = (\alpha_1^*\omega)(\mathbf{y})(g_*(\mathbf{x}; \mathbf{a}_i))_i,$$

with $p' = \alpha_0(\mathbf{x})$; the right side is $(g^*(\alpha_1^*\omega))(\mathbf{x})((\mathbf{x}; \mathbf{a}_i))_i$ by Definition 32.1.

Step 3. (d commutes with g^*.) On the interior of W, Theorem 32.3 and Step 2 give $d(\alpha_0^*\omega) = d(g^*(\alpha_1^*\omega)) = g^*(d(\alpha_1^*\omega))$. Both sides are continuous on W, and every point of W is a limit of interior points of W (if W is open in $\mathbb{H}^k$, points of $W \cap (\mathbb{R}^{k-1} \times 0)$ are limits of points of W with $x_k > 0$); so the identity holds on all of W.

Step 4. (Conclusion.) Let $\mathbf{v}_1, \dots, \mathbf{v}_{l+1} \in \mathcal{T}_p(M)$, $\mathbf{a}_i = \mathbf{v}_i(\alpha_0)$ and $\mathbf{b}_i = \mathbf{v}_i(\alpha_1) = Dg(\mathbf{x}_0) \cdot \mathbf{a}_i$. By Step 3 at $\mathbf{x}_0$,

$$\begin{aligned} d(\alpha_0^*\omega)(\mathbf{x}_0)((\mathbf{x}_0; \mathbf{a}_i))_i &= d(\alpha_1^*\omega)(\mathbf{x}_1)(g_*(\mathbf{x}_0; \mathbf{a}_i))_i \\ &= d(\alpha_1^*\omega)(\mathbf{x}_1)((\mathbf{x}_1; \mathbf{b}_i))_i. \end{aligned}$$

The two sides are the values of $d\omega(p)(\mathbf{v}_1, \dots, \mathbf{v}_{l+1})$ computed with α_0 and with α_1, so they agree. ■

Exercise 41.5 (Riemannian metrics exist)

Show that any compact differentiable manifold M has a Riemannian metric.

Solution. Here an inner product $\langle \cdot, \cdot \rangle_p$ on each $\mathcal{T}_p(M)$ is *of class* C^∞ if for every patch $\alpha : U \to V$ the functions $G_{rs}(\mathbf{x}) = \langle \alpha_*(\mathbf{x}; \mathbf{e}_r), \alpha_*(\mathbf{x}; \mathbf{e}_s) \rangle_{\alpha(\mathbf{x})}$ are C^∞ on U.

Step 1. (Local metrics.) Cover M by patches and let $\phi_1, \dots, \phi_l$ be as in Theorem 41.1, with Support ϕ_i covered by the patch $\alpha_i : U_i \to V_i$. For $p \in V_i$ and $\mathbf{v}, \mathbf{w} \in \mathcal{T}_p(M)$ let $\langle \mathbf{v}, \mathbf{w} \rangle_i = \mathbf{v}(\alpha_i) \cdot \mathbf{w}(\alpha_i)$ (dot product of components). Since $\mathbf{v} \mapsto \mathbf{v}(\alpha_i)$ is a linear isomorphism onto $\mathbb{R}^k$, this is an inner product on $\mathcal{T}_p(M)$.

Step 2. (Gluing.) Define

$$\langle \mathbf{v}, \mathbf{w} \rangle_p = \sum_{i=1}^{l} \phi_i(p) \langle \mathbf{v}, \mathbf{w} \rangle_i,$$

where a term with $\phi_i(p) = 0$ is read as 0 (so only patches with $p \in$ Support $\phi_i \subset V_i$ contribute). Each term is symmetric and bilinear, hence so is the sum. Since $\phi_i \geq 0$ and $\sum_i \phi_i(p) = 1$, some $\phi_j(p) > 0$; then for $\mathbf{v} \neq \mathbf{0}$,

$$\langle \mathbf{v}, \mathbf{v} \rangle_p \geq \phi_j(p) \|\mathbf{v}(\alpha_j)\|^2 > 0.$$

So $\langle \cdot, \cdot \rangle_p$ is an inner product on $\mathcal{T}_p(M)$.

Step 3. (Smoothness.) Let $\alpha : U \to V$ be a patch and $g_i = \alpha_i^{-1} \circ \alpha$ on $U_i' = \alpha^{-1}(V \cap V_i)$. By $(*)$ of Definition 41.4, $\alpha_*(\mathbf{x}; \mathbf{e}_r)(\alpha_i) = Dg_i(\mathbf{x}) \cdot \mathbf{e}_r$, so the ith term of G_{rs} is

$$h_i(\mathbf{x}) = \phi_i(\alpha(\mathbf{x})) \, (Dg_i(\mathbf{x})\mathbf{e}_r) \cdot (Dg_i(\mathbf{x})\mathbf{e}_s) \quad (\mathbf{x} \in U_i'), \qquad h_i(\mathbf{x}) = 0 \quad \text{otherwise.}$$

h_i is C^∞ on the open set U_i' ($\phi_i \circ \alpha$ and g_i are C^∞), and $h_i = 0$ on the open set $\alpha^{-1}(V - \text{Support } \phi_i)$; these two open sets cover U because Support $\phi_i \subset V_i$. Hence h_i and $G_{rs} = \sum_i h_i$ are C^∞ on U. ■

Exercise 41.6 (tangent vectors as derivations)

Let p be a point of $M - \partial M$. Let X be an operator on C^∞ real-valued functions defined near p satisfying locality, linearity and the product rule. Show there is exactly one $\mathbf{v} \in \mathcal{T}_p(M)$ with $X = X_{\mathbf{v}}$, as follows.
(a) Let F be C^∞ on the open cube $U = \{\mathbf{x} : |\mathbf{x}| < \epsilon\}$ in $\mathbb{R}^k$. Show there are C^∞ functions $g_1, \dots, g_k$ on U with $F(\mathbf{x}) - F(\mathbf{0}) = \sum_j x_j g_j(\mathbf{x})$.
(b) Show that $D_j F(\mathbf{0}) = g_j(\mathbf{0})$.
(c) Show that $X(c) = 0$ if c is a constant function.
(d) Show that there is at most one $\mathbf{v}$ with $X = X_{\mathbf{v}}$.
(e) Show that there is a $\mathbf{v}$ with $X = X_{\mathbf{v}}$.

Solution.

(a) Let

$$g_j(\mathbf{x}) = \int_{u=0}^{u=1} D_j F(x_1, \dots, x_{j-1}, u x_j, 0, \dots, 0).$$

Step 1. The point $(x_1, \dots, x_{j-1}, u x_j, 0, \dots, 0)$ lies in U for $\mathbf{x} \in U$, $u \in [0,1]$, and the integrand is a C^∞ function of $(\mathbf{x}, u)$. Applying Leibniz's rule (Theorem 39.1) on $Q \times [0,1]$ for closed cubes $Q \subset U$, repeatedly, every partial derivative of g_j of every order exists and is continuous; so g_j is C^∞.

Step 2. Substituting $t = u x_j$ (both sides are 0 if $x_j = 0$) and using the fundamental theorem of calculus,

$$\begin{aligned} x_j g_j(\mathbf{x}) &= \int_{t=0}^{t=x_j} D_j F(x_1, \dots, x_{j-1}, t, 0, \dots, 0) \\ &= F(x_1, \dots, x_j, 0, \dots, 0) - F(x_1, \dots, x_{j-1}, 0, \dots, 0). \end{aligned}$$

Step 3. Summing over $j = 1, \dots, k$, the right sides telescope to $F(\mathbf{x}) - F(\mathbf{0})$.

(b) At $\mathbf{x} = \mathbf{0}$ the integrand is the constant $D_j F(\mathbf{0})$, so $g_j(\mathbf{0}) = D_j F(\mathbf{0})$.

(c) By the product rule, $X(1) = X(1 \cdot 1) = X(1) \cdot 1 + 1 \cdot X(1) = 2X(1)$, so $X(1) = 0$; by linearity $X(c) = cX(1) = 0$.

(d) Let $\alpha : U \to V$ be a patch with U open in $\mathbb{R}^k$ and $\alpha(\mathbf{0}) = p$ (translate the domain if necessary), and let $h = \alpha^{-1} : V \to U$ with components $h_1, \dots, h_k$. Each h_i is C^∞ near p, since $h_i \circ \alpha(\mathbf{x}) = x_i$. If $X = X_{\mathbf{v}}$, then

$$X(h_i) = D(h_i \circ \alpha)(\mathbf{0}) \cdot \mathbf{v}(\alpha) = \mathbf{e}_i^{\mathsf{T}} \mathbf{v}(\alpha) = v_i(\alpha).$$

So the components of $\mathbf{v}(\alpha)$ are the numbers $X(h_i)$, and $\mathbf{v}$ is determined by X.

(e)

Step 1. With α, h as in (d), let $\mathbf{v}$ be the tangent vector at p with $\mathbf{v}(\alpha) = (X(h_1), \dots, X(h_k))$. Let f be C^∞ near p and $F = f \circ \alpha$, which is C^∞ near $\mathbf{0}$. Choose $\epsilon > 0$ so that the cube $U_\epsilon = \{|\mathbf{x}| < \epsilon\}$ lies in U and $\alpha(U_\epsilon)$ lies in the domain of f. By (a), $F(\mathbf{x}) = \sum_j x_j g_j(\mathbf{x}) + F(\mathbf{0})$ on U_ϵ.

Step 2. Since α is a homeomorphism onto the open set V, $\alpha(U_\epsilon)$ is a neighborhood of p, and on it

$$f = F \circ h = \sum_j h_j \cdot (g_j \circ h) + F(\mathbf{0}),$$

where $g_j \circ h$ is C^∞ near p because $(g_j \circ h) \circ \alpha = g_j$.

Step 3. By locality, linearity, the product rule and (c),

$$X(f) = \sum_j \left[X(h_j)\, g_j(h(p)) + h_j(p)\, X(g_j \circ h) \right] + X(F(\mathbf{0})).$$

Here $h(p) = \mathbf{0}$, so $h_j(p) = 0$, and $g_j(\mathbf{0}) = D_jF(\mathbf{0})$ by (b). Hence

$$X(f) = \sum_j D_jF(\mathbf{0})\, X(h_j) = D(f \circ \alpha)(\mathbf{0}) \cdot \mathbf{v}(\alpha) = X_{\mathbf{v}}(f).$$

■